

Engineering 110: Final Report

Danish Sain, Vikram Jangid, Gus Geremia, Manya Goutam

Problem Definition and Background

The CORAL after-school program is a Catholic charity organization that partners with local schools to provide students with hands-on STEM education through community-based projects. Each year, students program and assemble LED light ornaments that are displayed at Christmas in the Park, a widely attended community event that celebrates both the holiday season and the students' technical work. The display serves as a culminating showcase of student learning, making its reliable operation central to CORAL's educational mission.

The existing LED network, however, suffers from two recurring technical failures that undermine this mission. Individual nodes lose connectivity unpredictably during operation, and RF signal congestion causes widespread disconnection across ornaments mid-display. A third concern is environmental: winter conditions including rain and cold temperatures degrade ornament performance without proper weatherproof enclosures. Collectively, these failures require staff to spend significant time each year troubleshooting the system, diverting attention and resources away from student programming and event coordination. For an organization whose primary focus is youth education, this operational burden represents both a financial and mission-level cost. A more reliable, wireless solution is needed to ensure the display consistently reflects the work students have put into it and continues to serve as a meaningful community event year after year.

Review of the Field

The technical direction for this project was largely informed by our community partner, John, whose prior experience with the CORAL LED display system provided a practical foundation for our design decisions. The following review validates those recommendations through existing literature and establishes the technical context for our approach.

Wireless Communication: ESP-NOW

Our community partner recommended ESP-NOW as the wireless protocol for node communication based on his previous work with the system. ESP-NOW is a connectionless protocol developed by Espressif Systems that enables direct device-to-device communication at the MAC layer without requiring a router or access point [1]. This is particularly advantageous in dense RF environments like an outdoor public event, where standard WiFi networks suffer from congestion and latency issues. ESP-NOW maintains stable throughput with latencies under 10ms, making it well suited for a synchronized multi-node LED display where packet loss would produce visible inconsistencies [1].

Power Management: NeoPixel Arrays

John flagged power management as a critical design constraint early in our partnership, specifically the limitations of daisy-chained power delivery. This is well documented in the literature. WS2812B LEDs (NeoPixels) can draw up to 60mA per LED at full brightness, meaning 9 LEDs per ornament represents over 500mA of potential draw per node [2]. Across 8 ornaments this approaches 4A total, not accounting for microcontroller overhead. Voltage drop along a daisy chain is a known failure mode, with distal nodes receiving insufficient voltage and exhibiting color inaccuracies or dropouts [2]. Best practice recommends power injection at multiple points and current-limiting resistors on data lines [2].

PCB Design

Single-sided component placement was recommended by John to simplify the soldering process and reduce the risk of heat-induced component displacement during assembly. This aligns with standard PCB manufacturing guidance for low-volume hand-assembled boards, where double-sided population significantly increases assembly complexity and rework risk [3].

Weatherproof Enclosure

John emphasized weatherproofing as a non-negotiable requirement given the outdoor winter environment. The IEC 60529 ingress protection standard classifies enclosure sealing against water and particulates, with IP65 representing a practical minimum for rain-exposed outdoor electronics [4]. For low-volume production, 3D printed enclosures using weather-resistant filaments such as PETG or ASA provide adequate UV and temperature stability [5]. Minimizing unique part counts and integrating sealed cable entry points are established best practices that informed our enclosure design decisions [5].

Community Partner, User Requirements, and Project Specifications

The community partner for this project is the CORAL after-school program, a Catholic charity organization that works with local schools to provide students with hands-on STEM education. The Christmas in the Park display is a culminating event where students showcase their work to families and the broader community, making its reliable operation directly tied to CORAL's mission and reputation. When the display fails mid-event, it undermines the students' efforts and disappoints the families who attend specifically to see their children's work.

Through partner interviews and review of existing system documentation provided by John, the team identified three core user requirements. First, the system must maintain reliable wireless connectivity across all LED nodes with zero unplanned dropouts during the display. Second, the PCB must be single-layer to simplify hand assembly and reduce manufacturing complexity for a volunteer-driven organization with limited technical resources. Third, the total build cost must remain within a \$100 budget to remain feasible for a nonprofit operating on limited funding.

The value proposition of this project is a reliable, low-cost, weather-sealed wireless LED display system that eliminates the annual troubleshooting burden on CORAL staff, allowing them to redirect time and resources toward student programming rather than hardware maintenance.

Success for this project is defined by three measurable metrics: 100% node uptime during a full simulated display cycle, PCB dimensions fitting within the existing mounting hardware constraints, and a total bill of materials at or under the \$100 budget. These metrics were chosen because they map directly to the partner's operational pain points and provide objective criteria for validating the design.

Report Overview

Through our prototyping and planning we have encountered roadblocks, design restrictions, and decision structure which shape the path forward. While we will be unable to deliver a functioning network, we can hopefully lay the groundwork & current prototypes based on best-practice construction and resource use for the project to be continued in the future.

Report Roadmap

Given the constraints of a single academic quarter, the team focused on three core deliverables: scoping the system requirements through partner interviews and prior blueprints, designing a single-layer PCB in KiCad, and developing a weatherproof enclosure in CAD. While a fully functioning wireless network was beyond the scope of this quarter, the work completed establishes a validated foundation for future implementation.

Methodology-Prototyping

The team's prototyping strategy was shaped by three primary constraints: a \$100 budget, a single academic quarter timeline, and the need for a solution maintainable by a volunteer-driven nonprofit organization. Given these constraints, the team prioritized software-based design and validation over physical prototyping, focusing efforts on producing manufacturable design files that could be handed off for future physical implementation.

Tool selection centered on KiCad 10.0 for PCB design and CAD software for enclosure development. KiCad was chosen for its open-source accessibility and robust design rule checking capabilities, which allowed the team to validate designs without incurring fabrication costs. Team responsibilities were divided based on technical expertise. Vikram and Danish led technical development and documentation, overseeing PCB schematic design, layout, and report writing. Manya led software planning for the LED control system. Gus managed budget tracking, communications with the community partner, and mechanical enclosure considerations.

Design of Experiments (DoE)

The goal of the design validation experiment was to verify that the PCB design met both electrical correctness and physical manufacturability requirements before committing to

fabrication. Two formal checks were conducted within KiCad. The Electrical Rules Check (ERC) was used to identify schematic-level errors such as unconnected pins, missing power flags, and conflicting net assignments. The Design Rules Check (DRC) was then applied to the PCB layout to verify that trace widths, pad clearances, and board dimensions conformed to standard manufacturing tolerances and the partner's physical constraints. Any flagged errors were systematically reviewed and resolved before the design was considered complete. This iterative check-and-revise process served as the primary data collection and analysis method, with a clean ERC and DRC report representing successful validation of the design.

Results

1. PCB

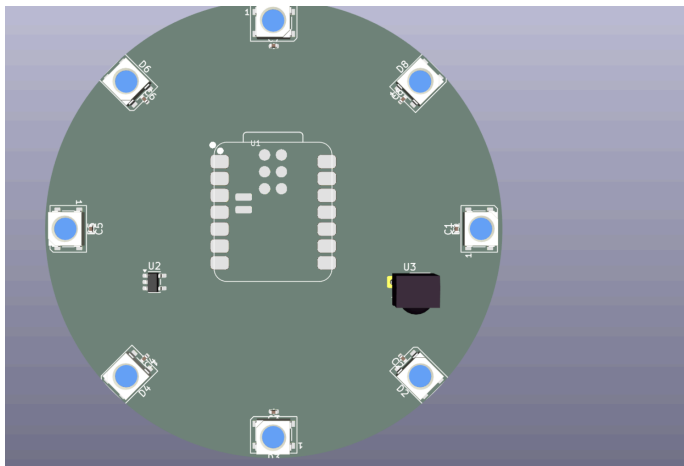
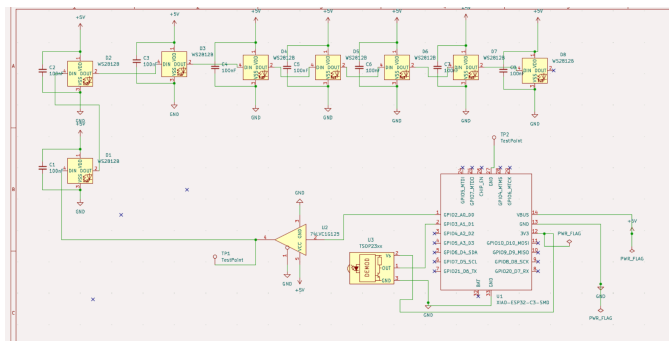
The primary deliverables of the prototyping phase were a validated PCB design and a bill of materials for the ornament node system. Figure 1 shows the KiCad 3D render of the completed PCB layout, featuring 8 WS2812B NeoPixel LEDs arranged radially around the perimeter of a circular board, with the ESP32-C3 microcontroller and supporting components centered on the board. The circular form factor was chosen to match the ornament mounting hardware and maximize LED visibility across the display. Figure 2 shows the corresponding schematic, which includes the ESP32-C3 module, a 3.3V to 5V level shifter on the NeoPixel data line, decoupling capacitors placed per LED for power stability, an AMS1117 LDO voltage regulator, and a reverse polarity protection diode on the power input.

The design was validated using KiCad's built-in Electrical Rules Check (ERC), which passed with no errors, confirming schematic-level correctness including proper net connectivity and

power flag assignments. The Design Rules Check (DRC) was also run to verify physical layout compliance with manufacturing tolerances.

The bill of materials for a single ornament node totals approximately \$5.56 in component cost, with the ESP32-C3-MINI-1U module representing the largest cost at \$5.00 per unit. Across 30 boards the total BOM cost comes to \$191.72, which exceeds the \$100 single-display budget.

However, the team only physically procured the ESP32-C3 modules this quarter as a proof-of-concept investment, with remaining components deferred to future procurement.



2. Enclosure

Using these Parameters - we settled on a single printable part to seat a 1/16th inch o-ring within a square groove. Through our research and partner review - a fully printed transparent

enclosure was ruled out. Because the project predicates on CORAL participants being able to see clearly their pattern displayed on the tree - even minimal diffusion should be avoided, ruling out PETG for the 'transparent' parts. In order to minimize printable pieces, (for both efficiency and cost) we decided - with some guidance from John - to utilize commercially available injection molded transparent ornament enclosures - carrying forward the printed o-ring seat for waterproofing. The injection molded pieces are more transparent than any printed pieces would achieve.

The printed part is identical for both sides of 'clamshell' - and clamps the o-ring between it with around 25% compression for a water tight seal. Axial compression is achieved with 6 M2 holes spaced around the enclosure for a bolt and nut configuration to achieve a seal within the face flange. The injection molded dome will be sealed into the closure mechanism with epoxy - the PCB will be similarly secured within the enclosure.

Below is a global and cross sectional view of the enclosure as it exists. Note the square groove - transparent clamshell.

Observations

The ERC passing cleanly indicates the schematic design is electrically sound at the logic level. The BOM coverage beyond the \$100 budget reflects the cost of scaling to 30 units and signals a need for future cost optimization, particularly through bulk ordering from LCSC and AliExpress as noted in the BOM. The removal of the IR sensor module and Header 3 from the design simplified the schematic and reduced cost compared to the prior v3.0 design, with QR code scanning via student tablet identified as a simpler activation method

Discussion

The ERC passing cleanly confirms the core electrical design is logically sound and ready for fabrication. The circular single-layer PCB layout satisfied the form factor and assembly requirements set by the partner. The BOM total of \$191.72 across 30 units exceeds the \$100 budget, primarily due to low-quantity pricing, and is expected to decrease significantly with bulk ordering. Of the three success metrics, PCB form factor was fully met. Node uptime could not be tested as ESP-NOW implementation was not completed this quarter. The budget metric was not met at scale, though the \$5.56 per-unit cost suggests feasibility with optimized procurement. The team's primary bottlenecks were unfamiliarity with KiCad, misaligned schedules, and a mid-quarter rescoping of the project. Rescoping away from a fully deployed network toward validated design artifacts was the right call, it allowed the team to deliver meaningful work within the quarter's constraints. John's domain expertise was critical in helping prioritize the right deliverables for project continuity.

Conclusion

This project addressed a recurring reliability problem in CORAL's annual Christmas in the Park LED display by designing a validated single-layer PCB and weatherproof enclosure system. While a fully deployed wireless network was beyond the scope of this quarter, the team delivered manufacturable design artifacts grounded in partner requirements and best-practice engineering principles. The ERC-validated schematic and circular PCB layout represent a concrete foundation for future implementation. The recommended next steps for a future team are to complete the ESP-NOW firmware implementation and conduct live node uptime testing, fabricate the PCB through LCSC or a similar low-cost manufacturer, 3D print and field-test the

weatherproof enclosure, and optimize the BOM through bulk ordering to bring per-display cost within the \$100 budget.

As a team, the most important lesson was the value of early scoping. Attempting to deliver a fully functional wireless network within a single quarter was unrealistic given our constraints, and rescoping mid-quarter allowed us to produce higher quality work within the time available. Communication and schedule alignment across team members remained a challenge throughout and is something future teams should establish clear systems for from the start. On a personal level, this project was an opportunity to learn KiCad from scratch, navigating schematic design, PCB layout, and design rule validation in a real engineering context made the experience genuinely rewarding despite the challenges.

References

[1] Espressif Systems, "ESP-NOW User Guide," Espressif Documentation, 2022. [online]

Available: <https://docs.espressif.com>

[2] Adafruit Industries, "NeoPixel Überguide," Adafruit Learning System, 2023. Available:

<https://learn.adafruit.com>

[3] ALLPCB, "The Ultimate Guide to Component Placement in Single-Sided PCB Assembly,"

ALLPCB Blog, Aug. 2025. Available:

<https://www.allpcb.com/blog/pcb-assembly/the-ultimate-guide-to-component-placement-in-single-sided-pcb-assembly.html>

[4] IEC, "IEC 60529: Degrees of Protection Provided by Enclosures," International Electrotechnical Commission, 2013.

[5] Sigma Filament, "An Expert's Guide to the Best Filament for Electronic Enclosures," SigmaFilament, Apr. 2026. <https://sigmafilament.com/best-filament-for-electronic-enclosures/>